Glasgow College, UESTC

Interim Report on UESTC4006P(BEng) Final Year Project

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Academic year	2026

Placement Company (if appropriate)	
Working Title of Project	AFDM transceiver design for underwater acoustic communication system
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Declaration of Originality and Submission Information	*I affirm that this submission is all my own work in accordance with the University of Glasgow Regulations and the School of Engineering requirements* Signed (Student) : Guo Yikun

1.Project Outline

Underwater acoustic (UWA) communication is characterized by a propagation environment that is significantly more challenging than terrestrial radio, primarily due to severe frequency-selective fading and significant time-varying effects. In UWA channels, the motion of transceivers or the medium introduces Doppler effects that are best represented as time-scaling (waveform compression/expansion) rather than simple frequency offsets. This results in extensive energy spreading in the delay-Doppler domain, making accurate channel acquisition and data recovery particularly difficult.

The objective of this project is to develop a robust simulation platform and a high-performance transceiver architecture for UWA communication using Affine Frequency Division Multiplexing (AFDM). The project employs a multi-scale, multi-lag (MSML) representation to model the UWA channel as a sparse set of dominant paths, each defined by a complex gain, propagation delay, and a Doppler scaling factor. This framework allows for the systematic evaluation of system performance across diverse acoustic scenarios by adjusting physical parameters such as delay spread and Doppler severity.

At this interim stage, the primary technical contribution is the design and implementation of an end-to-end AFDM link specifically optimized for UWA environments. While standard serial pipelines (utilizing OMP for estimation followed by MMSE or MP for detection) were established as initial baselines, the core innovation of this work lies in a proposed low-complexity joint channel estimation and detection algorithm. This scheme leverages the sparse delay-Doppler input-output relation of AFDM. It utilizes a "one-shot" OMP step to lock the delay-Doppler support, followed by an iterative loop that alternates between Message Passing (MP) for symbol inference and Least-Squares (LS) for gain refinement. This approach achieves superior performance by coupling the estimation and detection processes while maintaining near-linear computational complexity.

The performance of the proposed receiver is evaluated through Monte Carlo simulations, focusing on Bit Error Rate (BER) and Normalized Mean Square Error (NMSE) of the channel estimate. To isolate the impact of estimation errors, benchmarks including perfect-CSI references for both MMSE and MP detectors are provided. Furthermore, the system's reliability is analyzed under non-ideal conditions, addressing practical constraints such as synchronization requirements, pilot integrity under superimposed signaling, and the impact of Doppler-bin quantization.

2.Work done so far

2.1 MSML Underwater Acoustic Channel Modelling and Simulation

In this interim stage, the channel model is designed to capture the two dominant impairments in underwater acoustic (UWA) links—long multipath delay spread and Doppler effects that behave closer to

time scaling than a single frequency offset—while remaining controllable and reproducible for simulation. I adopt an MSML (multi-scale, multi-lag) view in which "lag" corresponds to multipath delays and "scale" corresponds to Doppler-induced distortions.

A standard starting point is a time-varying multipath impulse response in complex baseband form,

$$h(t,\tau)=\sum_{l=1}^{P}A_l(t)\delta(\tau-\tau_l(t)) \tag{2-1}$$

where each path l is described by a complex gain $A_l(t)$ and a time-varying delay $\tau_l(t)$. Over a sufficiently short processing block, the channel is assumed to vary slowly enough that gains can be treated as approximately constant, and the delay evolution can be approximated by a first-order model $\tau_l(t)\approx\tau_l-a_l t$, where a_l is a Doppler scaling factor (DSF). This highlights the MSML structure: τ_l captures the lag component, and a_l (or an equivalent scale factor) captures the scale component. When $a_l=0$ for all paths, the model reduces to a conventional delay-only multipath channel, which is useful for validation.

For simulation and receiver processing, the channel is realised in a discrete delay–Doppler form that matches the block-based baseband implementation. Each channel realisation is represented by a small set of dominant paths with (i) a discrete delay index ℓ_l (in samples), (ii) a Doppler-related parameter derived from the DSF, and (iii) a complex gain. Specifically, the DSF values $\{a_l\}$ are mapped to a normalised Doppler quantity

$$D_l=a_l\frac{f_c}{f_s} \tag{2-2}$$

and then quantised to an integer Doppler bin

$$k_l=\mathrm{round}(D_l N_{\mathrm{data}}) \tag{2-3}$$

where f_c is the carrier frequency, f_s is the sampling rate, and N_{data} is the AFDM block size after the guard segment. The Doppler bins are clipped to stay within bounds required by the AFDM parameter settings and orthogonality constraints. Together with the discrete delays $\{\ell_l\}$ and gains $\{A_l\}$, these parameters define a sparse delay–Doppler structure that is used to build the effective block channel matrix for AFDM processing.

The channel generation is controlled through a small set of parameters so that different regimes can be tested systematically. The number of active paths P is kept modest to reflect sparsity. Delays are generated within a prescribed delay-spread window, and complex gains are shaped by a power–delay profile so that later arrivals typically carry less energy. Doppler severity is controlled by the distribution of the underlying motion parameters (hence $\{a_l\}$), allowing experiments to sweep from mild to stronger

time variation. Reproducibility is ensured by fixed random seeds, logged parameter sets, and consistent Monte Carlo settings.

Validation is performed before using the channel module in higher-level system experiments. First, setting all scaling factors to zero ($a_l = 0$) should recover an ordinary multipath channel, verifying that delay and gain handling are correct. Second, a single dominant path with nonzero scaling should exhibit the expected time-scaling behaviour (observable through waveform alignment after delay compensation or through correlation-based checks), confirming that the scale component is present. Third, with multiple paths and mixed (ℓ_l, k_l), the received signal should display joint delay–Doppler spreading consistent with an MSML channel, for example through energy leakage across neighbouring delay/Doppler bins or broadened correlation features. These checks are presented with plots to make the verification transparent.

2.2 Complete Single-User AFDM System Implementation (Tx–Ch–Rx)

This section describes the implemented single-user AFDM link under the UWA channel and, in particular, the receiver structure used in the interim experiments. The simulation is organised around an AFDM-domain block model,

$$\mathbf{y} = \mathbf{H}_{\mathrm{DAFT}}(\boldsymbol{\theta})\mathbf{x} + \mathbf{w} \tag{2-4}$$

where $\mathbf{x} \in \mathbb{C}^{N_{\mathrm{data}}}$ is the transmitted block, $\mathbf{w}$ is complex Gaussian noise, and $\mathbf{H}_{\mathrm{DAFT}}(\boldsymbol{\theta})$ is an effective sparse channel matrix induced by a small number of dominant underwater paths. Each path is parameterised by a discrete delay ℓ_p, an integer Doppler bin k_p, and a complex gain g_p. In this representation, (ℓ_p, k_p) define the sparse delay–Doppler support, while $\{g_p\}$ determine the path strengths.

On the transmitter side, data symbols are generated and mapped to Gray-coded QPSK with unit average power. A pilot sequence of the same length is also generated and transmitted using superimposed pilots, so the block is formed as

$$\mathbf{x} = \sqrt{\alpha_d}\,\mathbf{x}_d + \sqrt{\alpha_p}\,\mathbf{x}_p \tag{2-5}$$

with fixed α_d, α_p in the interim setup. This choice keeps the block length unchanged and injects pilot energy across the entire frame, but it also means that during channel estimation the unknown data component appears as structured interference, which directly motivates the receiver design described below.

For each frame, the UWA channel is realised as a sparse set of paths with gains, delay indices, and Doppler-related factors. Delays are mapped to integer indices ℓ_p, and Doppler scaling is mapped to a normalised Doppler quantity and then quantised into integer Doppler bins k_p, with clipping to keep the bins within the guard region implied by the AFDM configuration. Given $\{(\ell_p, k_p, g_p)\}_{p=1}^{P}$ and the chosen chirp parameter c_1, the effective block channel $\mathbf{H}_{\mathrm{DAFT}}$ is constructed analytically. The guard segment length (CPP) is set to cover the maximum realised delay so that the usable processing length becomes

$N_{\text{data}} = N - \text{CPP}$, and feasibility checks tied to the delay–Doppler support are enforced during parameter setup to avoid inconsistent configurations.

At the receiver, several modes exist for benchmarking, but the main interim contribution is the joint channel estimation and detection scheme that improves performance over a purely serial pipeline while keeping complexity low. The receiver first uses OMP once to identify the dominant delay–Doppler support from pilots. After this step, the estimated delays and Doppler bins are frozen; only the path gains are refined thereafter. With support fixed, the received block can be viewed as linear in the gain vector: the observation is a sum of P path-specific responses weighted by $\{g_p\}$. The receiver then runs a short outer loop in which MP is used to infer the data symbols under the current channel estimate, and a small regularised least-squares (LS) update refines the gain vector using the current symbol estimate (data plus the known pilot component). This alternation is repeated a few times, after which final hard decisions are made by normalising the detected symbols and applying QPSK demodulation. The key design choice is that the iterative part only refines gains on a fixed, low-dimensional support; this captures most of the benefit of iteration without inflating the search space, and it naturally fits the superimposed-pilot setting where symbol uncertainty and gain estimation are coupled.

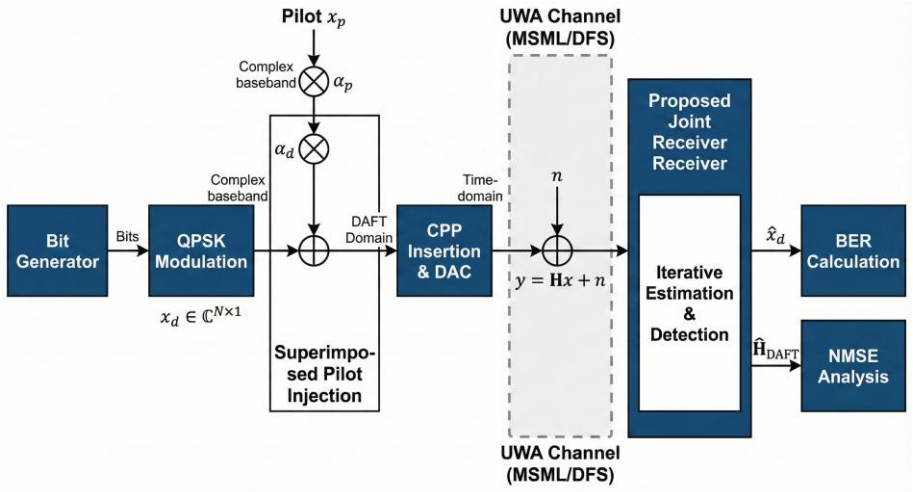


Figure 2-1. Single-user AFDM-UWA simulation chain (Tx–Ch–Rx) with superimposed pilots and the proposed joint receiver.

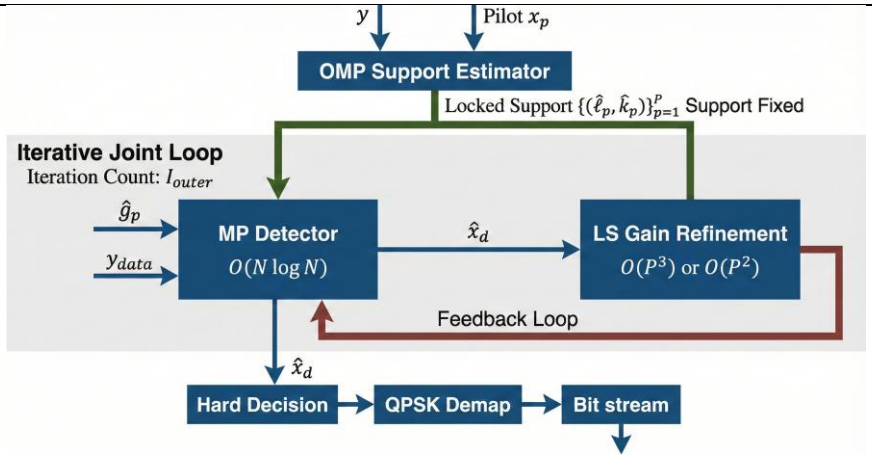


Figure 2-2. Proposed joint receiver: one-shot OMP support locking followed by MP detection and LS gain refinement in an outer iterative loop.

Table 2-1. Key simulation parameters for the AFDM-UWA experiments.

Parameter	Symbol	Value
Carrier frequency	fc	12 kHz
Bandwidth	B	4 kHz
Modulation	Mod	QPSK
Block length	N	1024
Number of paths	P	10
Max delay spread	τ_max	10 ms
Max Doppler shift	f_d,max	8 Hz
AFDM chirp parameter	c_1	0.0039
SNR range	SNR	0–25 dB

2.3 Channel Estimation and Detection Methods: Serial Baselines vs. the Proposed Joint Receiver

This section focuses on the key contribution of the interim work: a joint channel estimation and detection receiver tailored to the sparse AFDM input–output relation in the UWA MSML channel. Conventional serial pipelines first estimate the channel (e.g., via OMP) and then detect data using the fixed estimate, which makes BER highly sensitive to residual gain errors even when the delay/Doppler support is approximately correct. To address this, I propose a low-complexity joint scheme where OMP is used only once to lock the delay–Doppler support, and the receiver then iteratively refines the complex path gains while detecting the data. Specifically, MP detection and an LS-based gain update are alternated for a small number of outer iterations, while delays and Doppler bins remain fixed after the initial OMP step. This design preserves the sparsity-driven structure and avoids dense MMSE matrix inversions, keeping the overall complexity close to linear in block length under sparse connectivity. The effectiveness

of the joint refinement is evaluated using BER–SNR and NMSE–SNR curves under identical channel settings and pilot overhead.

The interim evaluation compares different receiver organisations under the same AFDM-UWA transmission settings, the same channel generator, and the same SNR grid. The purpose is not to compare waveforms, but to isolate the impact of (i) how channel knowledge is obtained and (ii) whether estimation is coupled to detection. Two serial baselines are first established. In the perfect-CSI baseline, the detector uses the true $\mathbf{H}_{\mathrm{DAFT}}$, which cleanly separates detector behaviour from estimation error and serves as an upper reference for the implemented detectors. In the one-shot OMP baseline, OMP is run once to estimate a sparse delay–Doppler channel (support and gains), an estimated $\mathbf{H}_{\mathrm{DAFT}}$ is reconstructed, and detection proceeds in a standard serial manner. This baseline is straightforward but can be sensitive to gain errors, especially when superimposed pilots are used and the estimator effectively sees data as interference.

The proposed receiver keeps the one-shot OMP step, but uses it only to lock the delay–Doppler support, treating $(\hat{\ell}_p, \hat{k}_p)$ as fixed thereafter. The receiver then couples detection and channel refinement through an iterative MP/LS loop: MP produces a symbol estimate given the current channel, and a low-dimensional LS step updates only the gain vector on the fixed support using the current reconstructed transmit block (data estimate plus known pilots). This is the central difference from the serial pipeline—rather than committing to a fixed $\mathbf{H}_{\mathrm{DAFT}}$ and detecting once, the receiver repeatedly aligns the gain estimate with the evolving symbol estimate. Empirically, this reduces BER relative to the serial OMP-based chain in the interim experiments, while preserving low complexity because the LS update operates on a P-dimensional parameter vector and MP exploits the sparsity pattern of $\mathbf{H}_{\mathrm{DAFT}}$.

Complexity considerations are reported alongside performance. The joint scheme is designed to remain close to linear scaling in the block length because its dominant costs are sparse-message updates (proportional to the number of retained edges after pruning) and LS updates on a small $P \times P$ system. This is in contrast to a direct MMSE detector that requires solving an $N_{\mathrm{data}} \times N_{\mathrm{data}}$ linear system, which becomes expensive as N_{data} grows. Performance is summarised with BER–SNR curves for the serial and joint receivers, and estimation quality is reported using the normalised Frobenius error between $\mathbf{H}_{\mathrm{DAFT}}$ and $\mathbf{H}_{\mathrm{DAFT}}$ (in dB). When reported, the post-refinement channel mismatch can also be included to show whether gain refinement reduces estimation error in parallel with BER improvement.

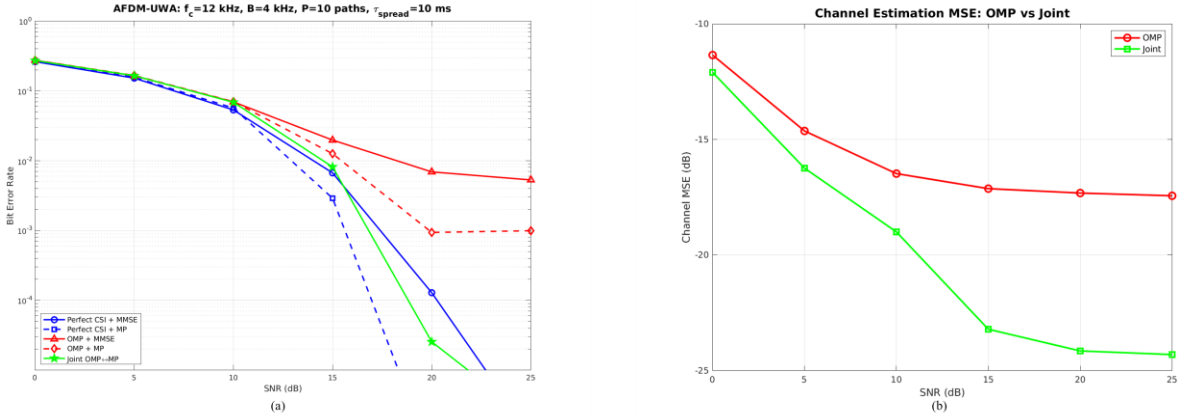


Figure 2-3. Performance under the MSML UWA channel: (a) BER versus SNR for serial baselines and the proposed joint receiver (with perfect-CSI references); (b) channel estimation NMSE versus SNR (one-shot OMP vs. proposed joint gain refinement).

Table 2-2. Complexity comparison of receiver pipelines (operation proxies): LMMSE+MMSE, OMP+MMSE, OMP+MP, and the proposed joint MP/LS refinement.

Method	Complexity (CMAC)
LMMSE + MMSE	$\mathcal{O}(NQ^2 + Q^3 + N^3)$
OMP + MMSE	$\mathcal{O}(PNQ + P^2N + N^3)$
OMP + MP	$\mathcal{O}(PNQ + I_{\mathrm{MP}} \cdot NP)$
Proposed Joint	$\mathcal{O}(PNQ + T(I_{\mathrm{MP}} \cdot NP + P^3))$

N: block length, $Q = (2K+1)(L+1)$: DD grid size, P: paths, I_{MP}: MP iterations, T: outer iterations

3. Challenges and Solutions

This section summarises the main technical challenges encountered in the interim implementation and explains how each issue has been handled in the current codebase, together with the next steps planned for verification and refinement.

3.1 Keeping AFDM parameters consistent with UWA delay–Doppler support

A first practical challenge is that underwater channels can exhibit both long delay spread and noticeable Doppler, and these directly constrain the AFDM parameter choices. In the implementation, the channel is realised through discrete delay taps and quantised Doppler bins, and AFDM must satisfy an orthogonality-related condition that depends on the effective delay support L, Doppler support K, and the usable block length N_{data}. If these supports are underestimated, the model becomes inconsistent (e.g., inter-symbol interference or loss of orthogonality can dominate); if they are overestimated too conservatively, spectral efficiency and complexity suffer.

What I did: Generating a “test channel” to estimate the maximum delay (to set the CPP length and therefore N_{data}) and to obtain a conservative Doppler bound. It then selects the chirp parameter c_1 with a

Doppler guard margin k_v, checks the orthogonality inequality, and automatically increases N if the condition fails. In addition, Doppler bins are clipped to remain within allowable bounds, preventing pathological cases where the quantised Doppler exceeds the AFDM configuration.

Next validation step: sweep the guard margin k_v and the Doppler statistics (mean/std velocity) to identify where performance becomes sensitive to the orthogonality margin, and report a simple threshold plot that shows when parameter choices start to break down.

3.2 Challenges in Error Propagation and the Performance-Complexity Trade-off

Initial system validation revealed that conventional serial estimation-detection pipelines—where channel state information (CSI) is estimated via OMP once and subsequently fixed—fail to effectively suppress the structured interference inherent in superimposed pilot schemes. In such setups, the unknown data symbols act as dynamic noise during the pilot-based estimation phase, leading to residual gain biases that inevitably propagate through the detection stage. While naive iterative methods could theoretically mitigate this, they often introduce new estimation errors or suffer from prohibitive computational costs (e.g., dense $N \times N$ matrix inversions in MMSE), making them unsuitable for resource-constrained underwater acoustic (UWA) platforms.

What I did: To address these limitations, I implemented a joint iterative receiver designed to achieve a robust trade-off between Bit Error Rate (BER) performance and computational overhead. The core innovation lies in a staged approach: Orthogonal Matching Pursuit (OMP) is employed exactly once to "lock" the sparse support of the UWA channel in the delay-Doppler domain. By fixing these coordinates, the receiver circumvents the high-dimensional dictionary search in subsequent iterations. The refinement process is then delegated to a low-complexity loop that alternates between Message Passing (MP) for symbol inference and a localized Least-Squares (LS) update for path gains. Since the LS update operates only on the P-dimensional path vector ($P \ll N$) and the MP detector exploits the sparse factor-graph structure of AFDM, the overall complexity remains near-linear with respect to the block length. This design effectively decouples the support identification from gain refinement, ensuring that symbol reliability improves without inflating the computational budget.

Next validation step: The next validation phase will focus on investigating algorithmic variants to further reduce the iteration count and exploring adaptive damping factors to enhance convergence stability. Additionally, I plan to discuss the scalability of this joint framework in high-mobility UWA scenarios where Doppler-bin quantization errors become more pronounced.

3.3 Off-grid mismatch from DSF-to-bin quantisation (delay–Doppler resolution limits)

Although Doppler in the physical model is continuous, the implemented block channel matrix relies on discretised Doppler bins $k_l = \mathrm{round}(D_l N_{\mathrm{data}})$. This quantisation is convenient and matches the sparse delay–Doppler representation used by OMP and MP, but it introduces an inherent modelling mismatch when true Doppler values lie between bins. A similar issue appears in delay if fractional delays are present but the implementation uses integer delay indices. The end result is an "off-grid" error floor that can limit estimation accuracy even at high SNR.

What I did: in the interim stage, I keep the simulator and estimator consistent by working on the same discrete grid and by restricting the OMP dictionary to a local delay–Doppler window derived from the realised channel support in each frame (which reduces ambiguity and improves stability). This makes the framework reliable for baseline studies and helps isolate the effect of other design choices (pilot strategy, detector selection, iteration).

Next validation step: introduce deliberate fractional offsets (or finer Doppler grids) and measure how quickly estimation error increases with grid mismatch. If off-grid effects are dominant, the next phase will consider dictionary refinement (local interpolation), finer search grids, or a lightweight post-OMP refinement step.

Following supervisor feedback, I separated on-grid validation from off-grid stress tests, so that implementation correctness is established before addressing grid mismatch.

3.4 Computational cost: dictionary growth in OMP and large-matrix operations

A recurring engineering challenge is runtime. OMP requires repeated correlations against a dictionary whose size grows with the delay–Doppler search window, and MMSE detection involves solving a linear system on an $N_{\mathrm{data}} \times N_{\mathrm{data}}$ matrix. MP can be more scalable on sparse matrices, but its cost depends on the number of retained edges and iteration count. With UWA settings (e.g., $N = 1024$, multiple paths, and Monte Carlo sweeps across SNR), naive implementations can become slow.

What I did: the implementation limits OMP search to a local window (using per-frame bounds), exploits sparsity in $\mathbf{H}_{\mathrm{DAFT}}$, and prunes small-magnitude edges in MP via a threshold. It also uses parallel execution (parfor) across SNR points and an adaptive early-stop rule: after a minimum number of frames, simulation at a given SNR stops once enough bit errors have been observed (with the triggering logic logged through the actual number of processed frames). These measures keep the interim experiments tractable without changing the underlying algorithms.

Next validation step: report a compact complexity table (dictionary size, OMP iterations, MP edges/iterations, and measured runtime per SNR), and evaluate whether iterative linear solvers or further sparsification can replace explicit MMSE matrix solves when N_{data} increases.

After discussion with my supervisor, I reported complexity using operation proxies (dictionary correlations, MP iterations, and LS updates) rather than only runtime, to make the comparison reproducible across platforms.

3.5 Security Considerations and Engineering Ethics

In accordance with the Engineering Council's definition of security, this project addresses security aspects from both operational and technical perspectives.

Firstly, as a software-based simulation study conducted in a controlled MATLAB environment, this project possesses inherently high operational security. Unlike physical underwater acoustic experiments that involve the deployment of high-power transducers or autonomous underwater vehicles (AUVs), this simulation-based approach eliminates risks related to maritime hardware loss, physical harm to marine ecosystems, or interference with national maritime infrastructure.

Secondly, from a technical standpoint, the robustness of the proposed joint receiver significantly enhances communication security. In the context of superimposed pilot signaling, the integrity of the pilot sequence is paramount for accurate channel acquisition. A potential security threat involves malicious "pilot spoofing" or structured interference designed to bias the channel estimate. By implementing a joint estimation and detection loop with LS-based gain refinement, the receiver demonstrates superior resilience against such structured noise compared to conventional serial baselines.

4. Conclusions from Initial Work and Future Work

4.1 Conclusions

The interim results presented in Figure 2-3 and the complexity analysis in Table 2-2 provide several critical insights into the performance of the AFDM-UWA transceiver framework.

First, the established MSML/DFS channel model and the corresponding AFDM transceiver pipeline have been validated as a consistent and reliable baseline for evaluating underwater communication performance. The simulation yields stable performance curves, confirming that the parameter mapping between physical underwater dynamics and the AFDM delay-Doppler grid is internally coherent.

Second, the Bit Error Rate (BER) analysis highlights the significant advantages of the proposed joint receiver, particularly in the high-SNR regime where noise is no longer the dominant limiting factor. As observed in Figure 2-3(b), conventional serial pipelines exhibit premature performance saturation. The OMP + MMSE baseline performs the worst, hitting an early error floor at approximately 8×10^{-3}, while the OMP + MP baseline also reaches a bottleneck around 1×10^{-3}. In sharp contrast, the proposed joint

iterative receiver shows no obvious error floor within the simulated range, effectively suppressing residual interference to reach a BER level of 10^{-5} at an SNR of 20 dB. This demonstrates that coupling symbol detection with gain refinement is essential for overcoming the estimation biases inherent in superimposed pilot schemes.

Third, the Channel Estimation Accuracy (NMSE) further supports this conclusion. While the stand-alone OMP estimator improves slowly and plateaus at approximately -17 dB, the joint refinement scheme converges rapidly, ultimately stabilizing at -24 dB. This 7 dB improvement in estimation quality directly translates to the observed BER gains.

Finally, the Complexity Analysis in Table 2-2 confirms that the proposed joint scheme maintains a highly favorable performance-complexity trade-off. By utilizing one-shot support locking and sparse-graph processing, the algorithm achieves near-linear complexity with respect to the block length N. This represents a substantial reduction in computational overhead compared to direct MMSE detectors, which scale cubically ($O(N^3)$), making the proposed architecture viable for real-time implementation on UWA platforms.

4.2 Future Work

In supervisor meetings, we agreed to prioritise a reliable single-user UWA AFDM baseline before moving to multi-user extensions. Based on this feedback, the interim work focuses on validating the MSML channel module and the proposed joint estimation–detection receiver, while multi-user modelling and multi-user detection are scheduled as the next-phase deliverables.

The next phase will focus on two concrete extensions: (i) introducing a multi-user UWA system model on top of the existing single-user chain, and (ii) developing and iterating detection methods that are suitable for multi-user interference in a delay–Doppler-sparse AFDM setting.

4.2.1 Multi-user system modelling (K-user superposition, synchrony control)

The current simulator will be extended from a single-user block model to a K-user uplink-style model in which each user has its own sparse delay–Doppler channel and its own transmit block. The received block will be the superposition of all users plus noise, with options to enable user-specific timing offsets and Doppler statistics. The main deliverable is not only the forward model, but also a set of controlled scenarios that can be repeated and stress-tested (e.g., equal-power users, near–far power imbalance, synchronous vs. asynchronous users, overlapping vs. separated supports in delay–Doppler).

Acceptance criteria: the model should (a) reduce exactly to the current single-user chain when $K = 1$, (b) generate reproducible K-user received blocks with configurable synchrony and power profiles, and (c) expose per-user ground-truth channels so that estimation and detection errors can be attributed correctly.

4.2.2 Multi-user detection algorithm development and iteration

With multi-user interference, a single "one-shot" detector is unlikely to be sufficient. The next step is therefore to build an iterative receiver that couples user separation, channel estimation, and symbol detection. A practical starting point is to leverage the existing ingredients (OMP and MP) in a multi-user loop: first obtain a coarse multi-user estimate (e.g., by extending the sparse dictionary to multiple users or by using block/group sparsity), then perform interference-aware detection (MP or linear detection on an expanded system), and finally refine the estimate through interference cancellation and re-estimation. Several design choices will be evaluated, such as whether to treat users jointly in one large factor graph or to use a staged approach (user detection → per-user decoding with successive interference cancellation).

Acceptance criteria: the receiver should demonstrate (a) measurable BER improvement over a simple linear multi-user baseline under the same simulation settings, (b) stable behaviour across a range of user counts and power imbalances, and (c) a clear complexity–performance trade-off reported via dictionary size, iteration counts, sparsity thresholds, and runtime. Where possible, genie-aided baselines (e.g., perfect CSI or known user activity) will be included to isolate which part of the pipeline limits performance.

5. Revised Gantt Chart

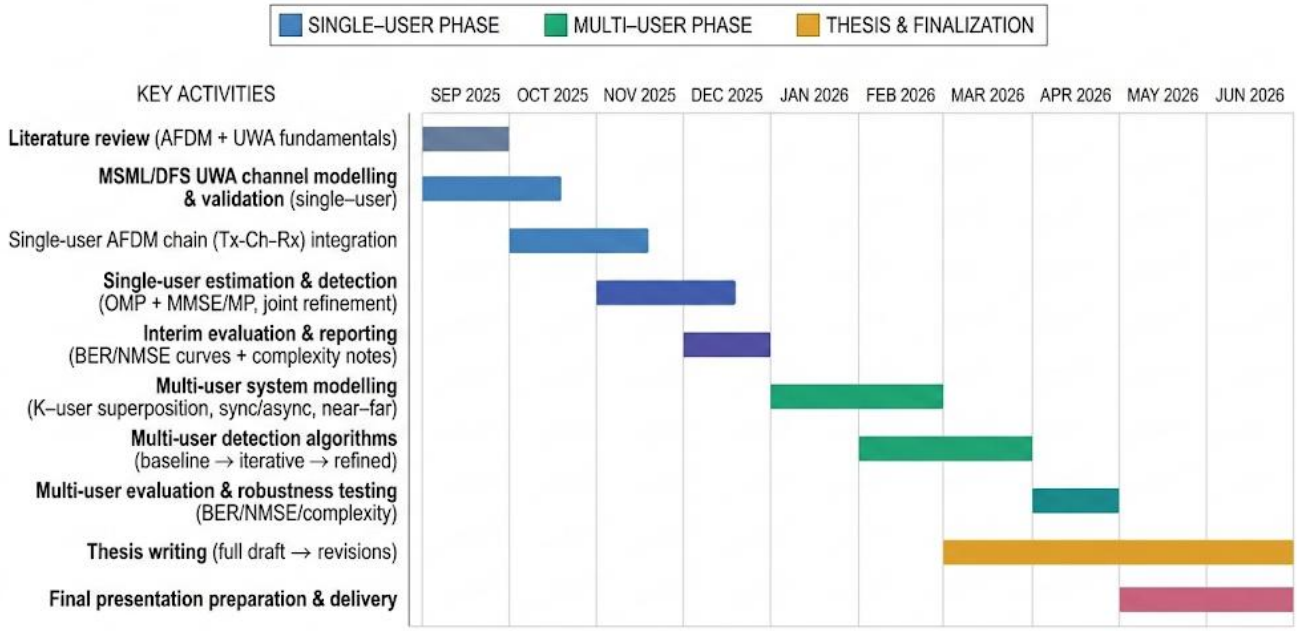


Figure 5-1. Revised project timeline (Gantt chart) highlighting completed single-user work and the planned multi-user phase.

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